



**Faculty of Engineering and Technology
Department of Electrical and Computer Engineering**

**Wireless and Mobile Networks
ENCS5323**

**Project
Real-World Spectrum Sensing and Controlled Radio
Frequency Analysis Using Software Defined Radio**

Instructor: Dr. Mohammad K. Jubran

Group Members:

Jumana Ajaj 1201768 Rahmaa 1212522

Shahd shreteh 1210444 Raghad buzia 1212214

Date: 11-7-2026

Introduction

Software Defined Radio (SDR) provides a flexible platform for analyzing real-world RF signals and studying wireless communication behavior beyond theoretical models. This project uses two ADALM-PLUTO SDR units to implement two complementary subsystems: a spectrum sensing system and a controlled interference generator.

The spectrum sensing subsystem analyzes RF activity in GSM 900, GSM 1800, and the 2.4 GHz ISM (Wi-Fi) band using Power Spectral Density (PSD) measurements and waterfall spectrograms to capture both occupied channels and their temporal behavior. The interference subsystem generates controlled in-band interference targeting the active Wi-Fi channel of a local access point, in order to quantify its effect on the performance of a connected wireless client — measuring round-trip latency, jitter, and packet loss under varying interference power levels. Together, the two subsystems demonstrate the relationship between spectrum activity, interference characteristics, and wireless network performance.

Part One: Spectrum Sensing

1.1 Method

The spectrum sensing system scans a selected frequency band by tuning the Pluto's local oscillator across overlapping windows, each corresponding to the SDR's instantaneous bandwidth. At each step, IQ samples are collected and processed using a Hanning-windowed FFT (1024 bins); multiple frames are averaged to reduce noise, and the resulting segments are stitched into one continuous PSD. Power is reported in relative dB (no absolute calibration stage), which is sufficient for identifying occupied channels and comparing relative signal strength. A waterfall mode repeats this measurement over time to visualize temporal spectrum variation.

1.2 Results and Interpretation

GSM 900 Downlink (935–960 MHz)

Eight carriers were detected above the occupancy threshold (-50.7 dB), against a noise floor of -56.7 dB. The strongest carrier (946.2 MHz, ≈ -18 dB/Hz) sits roughly 39 dB above the floor. The ~ 200 kHz spacing between carriers matches the GSM FDMA channel structure.

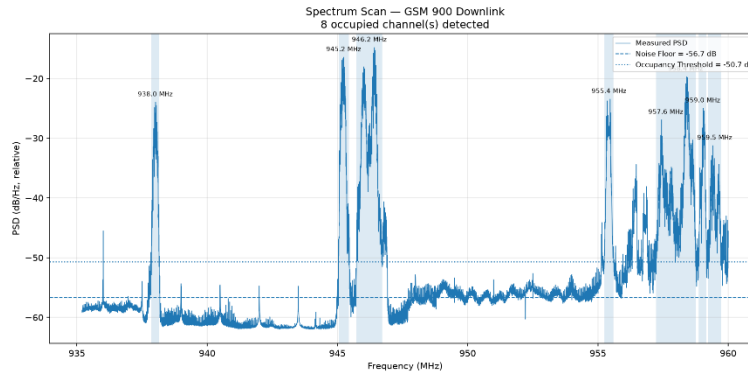


Figure 1: Spectrum Scan for GSM 900 Downlink (935–960 MHz)

GSM 1800 Downlink (1805–1880 MHz)

Several narrow peaks were observed, the strongest near 1814 MHz (≈ -20 to -22 dB), consistent with individual GSM carriers occupying distinct frequency slots.

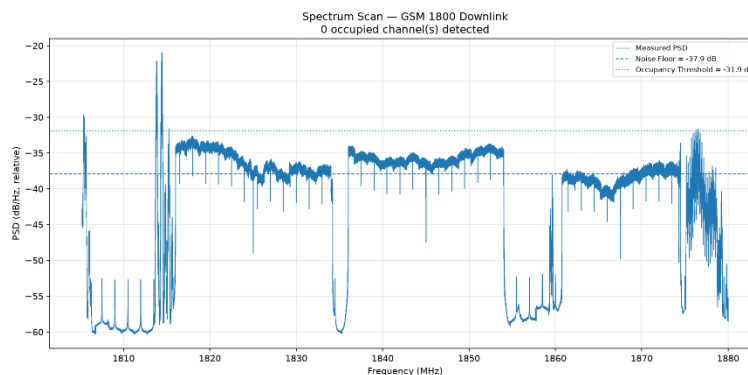


Figure 2: Spectrum Scan for GSM 1800 Downlink (1810–1880 MHz)

Wi-Fi Channels 1, 6, 11 (2.4 GHz ISM)

Channel 1 showed the strongest activity, with a clear occupied peak. Channel 6 showed a raised region (2432 – 2443 MHz) with two points above threshold and a burst-like pattern near 2442 – 2445 MHz, likely intermittent interference. Channel 11 remained close to its noise floor (-60.5 dB) with no detected occupancy. Since all three channels share the same 20 MHz bandwidth, this variation reflects differing local access-point/device density rather than any inherent channel difference.

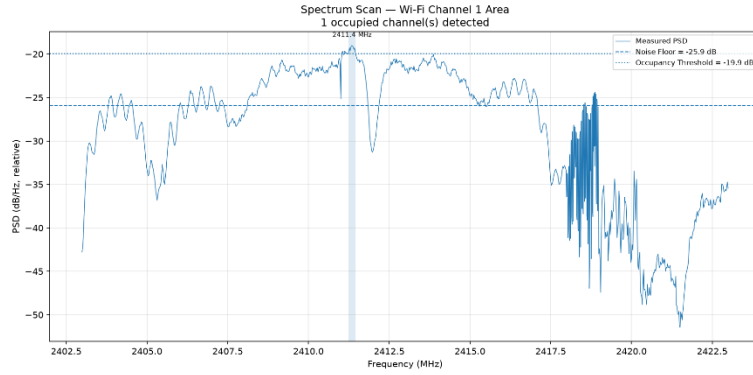


Figure 3: Spectrum Scan for Wi-Fi Channels 1 (2.4 GHz ISM)

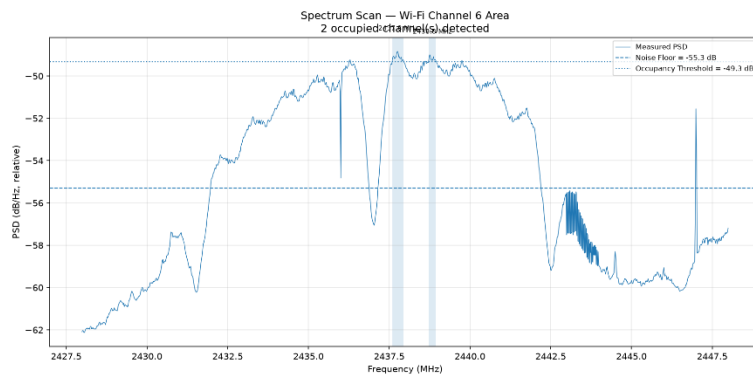


Figure 4: Figure 3: Spectrum Scan for Wi-Fi Channels 6 (2.4 GHz ISM)

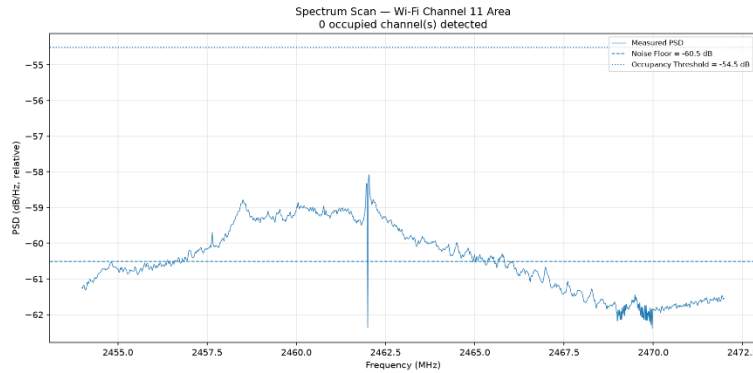


Figure 5: Wi-Fi Channels 11 (2.4 GHz ISM)

1.3 Temporal Variability

The waterfall shows GSM carriers as continuous vertical bands, confirming always-on transmission from the base station's broadcast channel. Wi-Fi, in contrast, appears as short bright bursts separated by idle periods — consistent with packet-based CSMA/CA access. The gap between Peak Hold (−10 to −15 dB) and Average PSD (−55 to −60 dB) on Wi-Fi channels

quantifies this: unlike GSM, the channel is occupied only intermittently, when a device actually transmits.

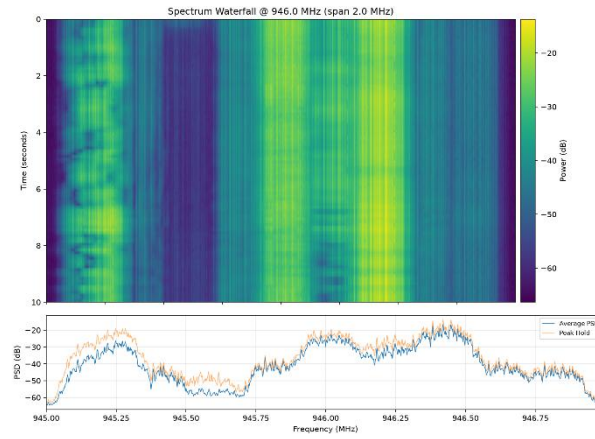


Figure 6: Waterfall for GSM 900 Downlink (946 MHz) [duration 10s]

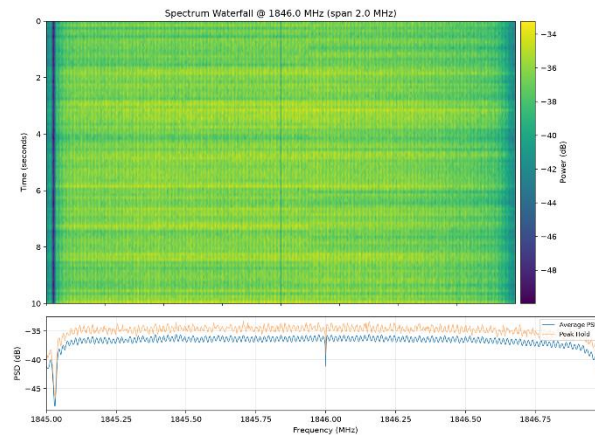


Figure 7: Waterfall for GSM 1800 Downlink (1846 MHz) [duration 10s]

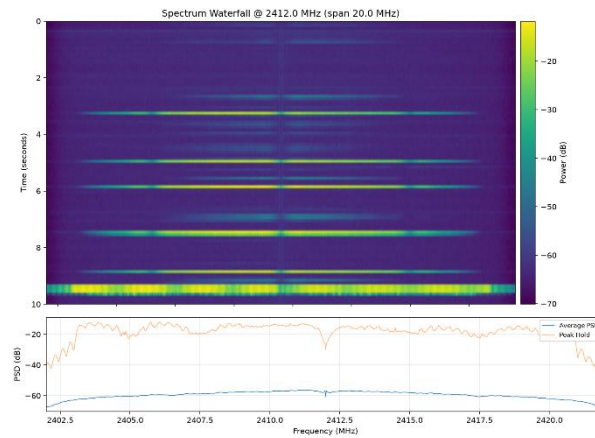


Figure 8: Waterfall for Wi-Fi Channel 1 Downlink (2412 MHz) [duration 10s]

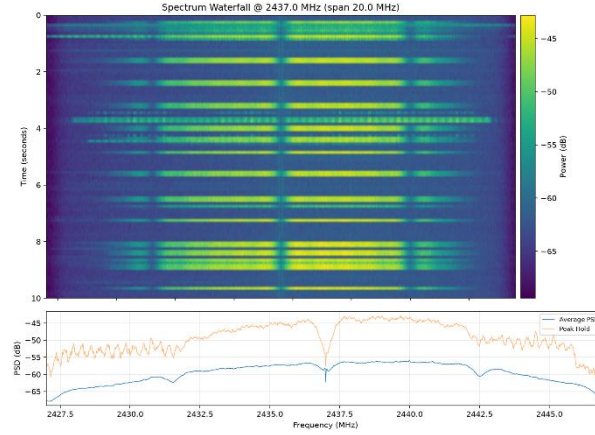


Figure 9: Waterfall for Wi-Fi Channel 6 Downlink (2437 MHz) [duration 10s]

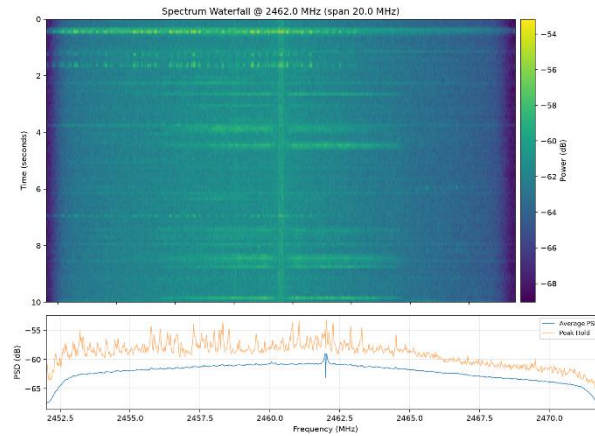


Figure 10: Waterfall for Wi-Fi Channel 11 Downlink (2462 MHz) [duration 10s]

Part Two: Controlled Interference Generation

2.1 Method

Pluto #2 transmits a controlled interference signal centered on the router's active Wi-Fi channel (2412 MHz), using a broadband noise waveform at adjustable TX power. A device connected to the access point is pinged repeatedly before, during, and after transmission, and round-trip latency and packet loss are compared across configurations to isolate the effect of the interference from normal network variation.

2.2 Interference Impact on Ping Latency

Spectral Evidence

A waterfall spectrogram recorded across the quiet and jammed intervals confirms the interference signal's presence on the target frequency: the noise floor rises by approximately 20 dB across the full 20 MHz channel span during transmission, visible as a clear brightening of the spectrum that begins and ends in step with the jammer's on/off timing.

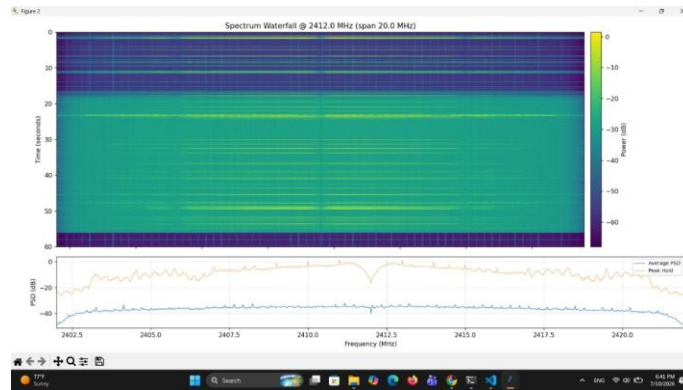


Figure 11: Waterfall spectrogram before (0–20s) and during (20–55s) interference transmission, showing the noise floor rising by approximately 20 dB across the full 20 MHz span.

Effect of Interference Power

Increasing the transmitted power produced a clear, monotonic degradation in link performance.

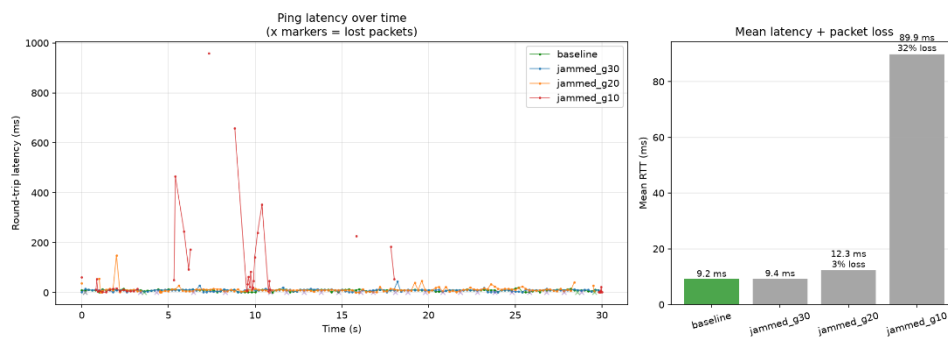


Figure 12: Mean latency and packet loss as a function of jamming power (TX gain)

Configuration	Mean RTT	Packet Loss
Baseline	9.2 ms	0%
Gain -30 dB	9.4 ms	0%
Gain -20 dB	12.3 ms	3%
Gain -10 dB	89.9 ms	32%

At the highest tested power (-10 dB), mean latency increased nearly $10\times$ and one-third of all packets were lost, demonstrating a direct dose-response relationship between interference power and receiver performance.

Effect of Waveform Type

At fixed power (-20 dB), the interference waveform shape changed the *nature* of the degradation rather than only its magnitude.

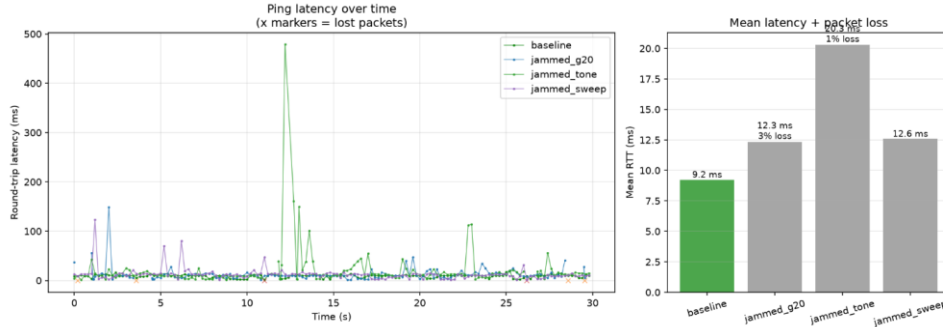


Figure 13: Effect of interference waveform type (noise vs. tone vs. swept CW) at fixed power (-20 dB).

Waveform	Mean RTT	Packet Loss
Baseline	9.2 ms	0%
Broadband noise	12.3 ms	3%
Narrowband tone	20.3 ms	1%
Swept CW	12.6 ms	0%

The narrowband tone produced the highest mean and peak latency (up to 479 ms) but the lowest packet loss, since its concentrated energy rarely collides with a transmission but when it does, the collision is severe. Broadband noise produced the opposite pattern: lower peak latency but the highest packet loss, since its energy is spread across the full channel, increasing the chance of colliding with some part of every transmission. This indicates a trade-off between interference concentration and frequency of impact, rather than one waveform being universally more disruptive.

Code Availability

The complete source code, along with detailed methodology, extended results, and full technical documentation, is available at:

[\[GitHub repository link\]](#)